

Roll No.

TPE-712

**B. TECH. (PETROLEUM ENGINEERING)
(SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

RESERVOIR MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the difference between Reservoir Engineering and Reservoir Management ? What is the various disciplines involved in Reservoir Management ? (CO1)
- (b) How does technology help in effective reservoir management ? What is the various technology required for reservoir characterization ? Explain in detail. (CO1)
- (c) What is reservoir mapping ? What are the various techniques required for Reservoir mapping and monitoring ? Explain each in detail. (CO1)

P. T. O.

2. (a) What are the *five* important different phases of Oil and Gas field ? Explain each in detail. Draw a cash flow diagram of an Oil and Gas field with time. (CO2)
- (b) What are the *three* different phases of development plan in view of recovery mechanism ? What is the basic condition for employing secondary and tertiary recovery mechanism ? (CO2)
- (c) Design a Field Development Plan (FDP) with long-term production forecasts to support the business strategy and goals. (CO2)
3. (a) Explain Petroleum Geology. What are the essential elements of Petroleum Geology ? Explain the terms Source, Reservoir, Seal, Trap, Maturation, Pool with figures. (CO3)
- (b) Explain primary, secondary porosity, facies, permeability, effective and relative permeability. Explain Darcy's law of permeability. (CO3)
- (c) f Explain Variogram with figure. How this is useful for petrophysical modeling process ? Write the formula for variance calculation. (CO3)
4. (a) Explain Depletion drive and Gas cap drive reservoir. Draw a comparison diagram of Pressure vs. Time of both type of reservoir. (CO4)
- (b) What is the common cause for water production from an oil and gas field ? What are the *two* different methods for water control ? Explain with diagram. (CO4)
- (c) Explain EOR and when it is applied in the field. What is the essential requirement for EOR application ? What are the different methods applied for achieving maximum oil from tertiary recovery ? (CO4)

(3)

5. (a) What are the *three* different categories of Reserve ? Explain each in detail in view of P90, P50 and P10. What are the different methods for reserve estimation ? (CO4 & CO5)
- (b) From logs, it has been determined that a productive formation has a bed thickness of 20 ft, a porosity of 25%, and a water saturation of 30%. The recovery factor is assumed to be 30%. The oil formation volume factor is 1.3. The areal extent of the reservoir is 200 acres. (CO5)
- Find :
- (i) HCPV expressed in ft
 - (ii) OIP expressed in bbl/acre
 - (iii) Reserves expressed in stock-tank barrels
- (c) What is the difference between Risk and Uncertainty ? What is the various factors responsible for Uncertainty in measurement ? What are the SPE guidelines for E&P Companies for probability of Exceedance and non-Exceedance for Oil and Gas assessment ? (CO5)

Roll No.

TPE-717

**B. TECH. PETROLEUM ENGINEERING
(SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

ADVANCED EOR TECHNIQUE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the classification of EOR as a thermal and non-thermal enhanced oil recovery ? (CO1 & CO2)
(b) Explain the different oil production phase ? Why we need EOR ? Currently available EOR is based on which parameter ? (CO1 & CO2)
(c) Explain the different oil production phase ? Why we need EOR ? What are the different techniques of EOR ? (CO3)
2. (a) Discuss the operational activities in pressure measurement during the production of crude oil ? (CO3 & CO4)

P. T. O.

- (b) Define static and bottom hole pressure ? Write and define the different types oil field pressure calculation requirements. (CO1 & CO4)
- (c) The data of depth and pressure is given in the below : (CO1 & CO2)

Depth (m)	Press. (atm)
0	51
300	55.3
600	59.5
900	63.6
1200	81.2
1500	99.0
1800	119.0
2100	141.0

Calculate the following :

- (i) Pressure gradients
- (ii) Depth of gas-oil and gas-water interface
- (iii) Pressures at GOC and WOC

Given : Depth of top perforation = 1140 m, GOC = 745 m, WOC = 1160 m.

3. (a) Define and derive Arp's equation and explain with suitable diagram ? What are the types of rate decline behavior ? Derive expressions for rate-time relationship for harmonic and exponential curve. (CO1)

(3)

- (b) Draw and explain with diagram of the effect of trapped gas on water flood recovery ? Discuss the different selection patterns of water flooding. (CO1 & CO2)
- (c) What are the typical displacement system in the miscible slug flooding ? Explain with the illustration of schematic diagram. (CO2 & CO5)
- 4. (a) Explain the ternary diagram. Discuss all the diagram by considering the different pressure and temperature along with the phase diagram inside the ternary diagram. (CO1 & CO5)
- (b) What are the basic methods of miscible drive ? Explain the phase conditions in the reservoir during high pressure gas injection. (CO4 & CO5)
- (c) Explain the following : (CO3 & CO6)
 - (i) LPG slug injection
 - (ii) Alcohol slug injection
- 5. (a) Discuss the Reservoir Selection and Process design during advanced enhanced oil recovery. (CO4 & CO6)
- (b) Draw the Schematic diagram of MEOR and explain the each steps in increase the production rate of microorganism during the enhanced oil recovery. (CO4 & CO5)
- (c) Explain Microbial Recovery Techniques ? Its Principles, Applications and Potential ? Discuss any projects for MEOR in Indian scenario. (CO2 & CO3)

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Roll No.

TPE-719

**B. TECH. (PETROLEUM ENGINEERING)
(SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023
CITY GAS DISTRIBUTION**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you understand by LNG ? With the help of a diagram, explain LNG Value Chain and write the names of the company which works for LNG. (CO1)
- (b) Explain what do you understand by PNG. Also describe how PNG is a clean energy for household and what is the basic difference between Commercial and Industrial PNG. (CO1)
- (c) Explain City gas distribution, Evolution of CGO in India and write *ten* names of CGD prominent companies in India. (CO1)
2. (a) Write short notes on the following : (CO2 & CO3)
 - (i) CNG

P. T. O.

(2)

(ii) Diaphragm Meters

(iii) Spur Pipeline

(b) With the help of a table, explain in brief Natural Gas composition.

(CO2 & CO3)

(c) Briefly explain Gas metering and Classify types of gas meters and describe any *two* gas meters.

(CO2 & CO3)

3. (a) Explain Distribution and Transmission Pipeline and also write classification of pipeline based on pressure gas to be transported.

(CO2, CO3 & CO4)

(b) On the basis of function, access and jurisdiction classify pipeline.

(CO2, CO3 & CO4)

(c) Explain district regulating station and gas odorization process.

(CO2, CO3 & CO4)

4. (a) Differentiate between Steel pipelines and Polyethylene pipelines.

(CO4)

(b) Explain city gas station and describe all city gas station operations.

(CO4)

(c) Describe safety guidelines for gas leakage.

(CO4)

5. (a) Explain what is gas leakage and describe how we can detect gas leakage.

(CO5)

(b) Write effects of gas leakage on environment and what different safety measure should be taken to avoid gas leakage.

(CO5)

(c) Explain in brief the Gas Allocation and Gas Availability issues in City Gas Distribution.

(CO5)

Roll No.

TPE-724

B. TECH. (PETROLEUM ENGINEERING)

(SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

PETROLEUM ECONOMICS, RISK AND UNCERTAINTY ANALYSIS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is risk ? Describe the different types of risk in petroleum industry. (CO1)
- (b) Discuss the importance of risk and uncertainty analysis in petroleum industry. (CO1)
- (c) What is exploration license ? Discuss the salient features of New Exploration License Policy (NELP) in India. (CO1)
2. (a) What is cash flow statement ? Discuss its significance. (CO1)
- (b) Explain cash flow from operating activities. (CO1)

P. T. O.

(2)

- (c) Explain how to ascertain the cash flow from investing and financing activities. (CO2)
- 3. (a) What is Net Present Value (NPV) ? Discuss the role of NPV in economics decision-making. (CO2)
- (b) What is the significance of internal rate of return in economic decision-making in petroleum industry ? (CO2)
- (c) What is capital productivity index ? Discuss. (CO2)
- 4. (a) What is royalty tax regime ? Discuss. (CO3)
- (b) What is production sharing contracts ? Discuss. (CO3)
- (c) List and describe the different types of fiscal regime in hydrocarbon exploration and production. (CO3)
- 5. (a) What are the geological risk and uncertainty involved in petroleum exploration. (CO4)
- (b) Discuss the Expected Monetary Value (EMV) analysis methods of quantifying risk. (CO5)
- (c) Discuss the managing risk through portfolio optimization. (CO5)